

**Client****Server**ClientHello:  $pk_{pq} \parallel pk_{tr}$  (1216 B)
$$\text{encapsulate} \Rightarrow ct_{pq}, ct_{tr}, ss_{pq}, ss_{tr}$$
$$K = \text{CONTEXTBOUND}(\dots)$$
ServerHello:  $ct_{pq} \parallel ct_{tr}$  (1120 B) $\{\text{Cert}, \text{CertVerify}, \text{Finished}\}_K$ 
$$\text{decapsulate} \Rightarrow ss_{pq}, ss_{tr}$$
$$K = \text{CONTEXTBOUND}(\dots); \text{verify cert}$$
 $\{\text{Finished}\}_K$ 

authenticated session key  $K$ ; KEX secrecy fails only if *both* KEM legs fail